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Journal of Marketing for Higher Education

ISSN: 0884-1241 (Print) 1540-7144 (Online) Journal homepage: https://www.tandfonline.com/loi/wmhe20

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To cite this article: Mollie Dollinger, Jason Lodge & Hamish Coates (2018) Co-creation in higher education: towards a conceptual model, Journal of Marketing for Higher Education, 28:2, 210-231, DOI: 10.1080/08841241.2018.1466756

To link to this article: https://doi.org/10.1080/08841241.2018.1466756

Published online: 27 Apr 2018.

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JOURNAL OF MARKETING FOR HIGHER EDUCATION
2018, VOL. 28, NO. 2, 210–231
https://doi.org/10.1080/08841241.2018.1466756

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Co-creation in higher education: towards a conceptual model

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ABSTRACT
Students have begun to show interest in adopting active and participatory roles that allow them to interact and work collaboratively with educators. One important aspect of students as partners is a process known as value co-creation. Value co-creation is the process of students' feedback, opinions, and other resources such as their intellectual capabilities and personalities, integrated alongside institutional resources, which can offer mutual value to both students and institutions. This paper presents the first conceptual model of value co-creation in higher education using a lens of co-creation cultivated through business and marketing literature. The model includes key components of value co-creation, co-production, and value-in-use as well as links to the anticipated benefits of value co-creation. The model can be used to inform and guide practice for the faculty and administration within higher education as well as to broaden the foundation of value co-creation literature.

ARTICLE HISTORY
Received 9 June 2017
Accepted 2 October 2017

KEYWORDS
Value co-creation; higher education; marketization; co-production; value-in-use

Introduction

Students and higher education institutions both have various incentives to improve the student experience. Institutions are motivated by their desire to find additional revenue, as government subsidies are reduced and global competition grows (Frasquet, Calderón, & Cervera, 2012; Mark, 2013). To attract and satisfy students, universities are applying concepts formerly confined to business, such as relationship marketing theories, that help facilitate organizations to build strong relationships with their consumers (Schlesinger, Cervera, & Iniesta, 2015). Students, meanwhile, have shown interest in playing a more active role in their higher education experience as partners (e.g. Bovill & Felten, 2016; Healey, Flint, & Harrington, 2014). Motivated by growing tuition fees and graduate employment concerns, students are looking to find value in their degrees that will satisfy their current and future needs (Tomlinson, 2008; Vickers & Bekhradnia, 2007). Yet, while both institutions and students are motivated to improve the student experience, there are few pathways to help them do so collaboratively.

The process of value co-creation can allow for institutions and students to work together to improve the student experience and enhance students' ability to act as partners. Value co-creation is a process whereby students' resources are integrated with

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organizational resources to facilitate a range of activities and experiences that encourage exchange and interaction which can lead to better practice and innovation (Prahalad & Ramaswamy, 2004). Examples of student resources include their intellectual abilities, study habits and methods, sense of responsibility and personality (Díaz-Méndez & Gummesson, 2012) as well as their perspectives and opinions on their learning experiences. However, value co-creation has just begun to be applied to higher education (e.g. Díaz-Méndez & Gummesson, 2012; Elsharnouby, 2015; Navarro-García, Peris-Ortiz, & Rueda-Armengot, 2015) and no comprehensive conceptual model of co-creation in higher education exists. Therefore, the first contribution of this paper will be to put forth a model to explore value co-creation in the higher education context.

The second contribution of this paper will be to add to the burgeoning literature that suggests that value co-creation contains two distinct dimensions. Ranjan and Read (2016), in a literature analysis of value co-creation, have found that the process actually comprises two distinct yet complementary dimensions: co-production and value-in-use. Yet Ranjan and Read (2016) also found that 79% of co-creation studies included in their dataset only considered one of these critical dimensions. Therefore, much of the research in value co-creation has not adequately acknowledged its complex underpinnings (Grönroos, 2011; Ranjan & Read, 2016). Value co-creation is a two-dimensional concept that includes two sub-processes: first, the co-production of value, where the value proposition is created alongside the consumer, and second, value beyond production into the consumption of such value, a concept from service-dominant logic known as value-in-use (Lusch & Vargo, 2006). For example, Droge, Stanko, and Pollitte (2010) use a co-creation framework to discuss user blogs for new product development (NPD); however, within their analysis they focus only on the co-production benefits to the organization without in-depth analysis of how the blog impacts consumers' perceived value (i.e. value beyond production) as well. Yet, as Vargo and Lusch (2011) point out, co-creation is not only a value proposition made to the organization through a co-production process, but also the value actors can make for themselves post-production. Therefore, despite some researchers confusing co-creation as equal to co-production, it is in fact made up of both co-production and value-in-use (Ranjan & Read, 2016). This paper will build from Ranjan and Read's (2016) analysis that links co-creation to both underlying constructs and apply a nuanced understanding of co-creation to the higher education sector by modifying both co-creation dimensions – co-production and value-in-use.

Finally, this paper will begin to link the process of value co-creation to the anticipated benefits that might create mutual value for institutions and students. Consistent with diffusion of innovation theory (Rogers, 2003), in order for value co-creation to gain the necessary traction and interest in research, there must first be a detailed argument for its anticipated benefits, not only to the student population, but also for institutional benefit.

Value co-creation

Higher education has for a long time been concerned with student engagement (e.g. Chickering & Gamson, 1999; Kuh, 2003). Discussion has pointed especially to student-staff partnerships as an important facilitator to deeper student engagement and participation (Bovill & Felten, 2016; Umbach & Wawrzynski, 2005). Outside of higher education, however, the trend of actively involving consumers within the organization has also been

growing (Ramaswamy & Ozcan, 2014). This paper will therefore attempt to merge existing co-creation literature to the current higher education context.

Value co-creation is most commonly known as a marketing and management strategy that incorporates consumer resources to jointly co-create new and innovative forms of value (Perks, Gruber, & Edvardsson, 2012). Value co-creation emerges from the breakdown of the traditional producer-consumer exchange, whereby the producer creates value and the consumer consumes or destroys it (Vargo & Lusch, 2008). Conversely, in the adoption of a co-creation process, the organization forms a reciprocal and balanced relationship with their consumers, allowing for a wider range of consumer voices and preferences to shape all aspects of the service, product, and/or delivery (Prahalad & Ramaswamy, 2004). Value co-creation further takes the perspective that it is unnecessary to distinguish products from services or vice versa, as all products contain service elements, and within all services there are product (otherwise known as 'goods') elements (Gummesson, 2007). Instead, the output of marketing actions can be thought of as 'value propositions' that interact with the consumer, thus making the consumer the co-creator of value (Gummesson, 2007). By deviating from traditional consumer/organizational roles, the producer no longer presumes to understand the needs of its consumers, and instead opens the formerly closed producer-process to allow for more consumer involvement. Consumers are allowed and encouraged through co-creation to share opinions, complain, negotiate, endorse, and interact with their organization in new ways (Cova & Dalli, 2009).

By allowing students and institutions to connect and collaborate, value co-creation bridges formerly isolated users' and producers' asymmetric knowledge and information, which has been shown to be a barrier between producers and consumers (von Hippel, 2009). Through co-creation, students' differing knowledge and resources can jointly interact with university faculty and staff to further create more integrated and superior outcomes than if only one group tried to satisfy the needs of the other alone (Frow, Nenonen, Payne, & Storbacka, 2015; von Hippel, 2009; Zwass, 2010). Already within the private sector, firms that utilize more consumer resources to jointly create value are shown to have more profits and larger market shares (Cefis & Marsili, 2005; Czarnitzki & Kraft, 2004).

Co-production and value-in-use

The following sections will supply an overview of the two underlying concepts of value co-creation – co-production and value-in-use. Within these sections, co-production and value-in-use will additionally be modified and applied to the higher education context through the formative indicators Ranjan and Read (2016) identified as critical to the underlying constructs of co-creation, and ultimately, to value co-creation.

Co-production

Co-production with consumers is a system allowing for consumer resource integration very early in the value chain of production, where consumers' knowledge, experiences and opinions can influence the creation of the service and product (i.e. value proposition) (Lusch & Vargo, 2006). While co-production is often misconstrued as customization, the latter is often just a superficial change to the service and production in later stages of

production (e.g. changing the initials on a pillowcase) (Tsigkas, 2013). Co-production differs from superficial processes such as customization because it occurs much earlier in the production stages, but also because the consumers' resource integration impacts many consumers of the product. In customization, often only the single consumer is impacted (Bovaird & Loeffler, 2013). Humphreys and Grayson (2008) therefore write that for an activity to constitute as co-production, there needs to be a fundamental shift in the production process, and consumers need to be engaged through design, knowledge production, or feedback that can deeply impact the value proposition.

The process of co-production rests on many of the same principles as co-creation, such as active participation with the consumer (Auh, Bell, McLeod, & Shih, 2007) and continuous dialogue that facilitates collaboration (Grönroos, 2011; Prahalad & Ramaswamy, 2004). Within co-production, consumers can be thought of as partial employees or part of the production team (Lengnick-Hall, 1996). Responsibility is also more equal than in a traditional producer-consumer exchange, as consumers are also asked to integrate knowledge and experiences and interact as part of the team with the organization (Mark, 2013).

Ranjan and Read (2016) write that co-creation is a third order construct, encompassing elements of both co-production and value-in-use, which further break down into three additional constructs. To illustrate, co-production can be thought of as the first stage of co-creation. It is where the consumer can influence and impact the organization's value proposition. The co-production interaction between the organization and consumer can further be assessed through three sub-processes or formative indicators: knowledge, equity, and interaction (Ranjan & Read, 2016). These indicators together form the construct of co-production, a composite variable. In the higher education context, we can modify these constructs to: (1) How does the student integrate their knowledge, experiences and/or other resources into the value proposition of higher education? (2) Does the student have equal access to the development and design of the higher education value proposition? (3) What is the quality of the interactions between the student and the higher education institution to integrate resources and co-create the value proposition? (Table 1)

To further illustrate how each of these constructs can be modified within the higher education context, this paper will discuss each of three constructs below and explore their application to higher education.

Knowledge

A foundational element to co-production is knowledge sharing, or how organizations and consumers combine their knowledge and skill sets to achieve greater innovation (Ramirez,

Table 1. Co-creation second order construct Part II: co-production.

Second-order construct	First-order construct	Modification to higher education
Co-production	Knowledge	How does the student integrate their knowledge, experiences and/or other resources into the value proposition of higher education?
	Equity	Does the student have equal access to the development and design of the higher education value proposition?
	Interaction	What is the quality of the interactions between the student and the higher education institution to integrate resources and co-create the value proposition?

1999). Knowledge sharing is achieved through various mechanisms to allow the organization to collect and analyze consumers' opinions and knowledge with or relating to the value proposition.

Sharing knowledge between consumers and organizations is a critical mechanism in the co-production process as it helps identify current and future needs (Gibbert, Leibold, & Probst, 2002). Gibbert et al. (2002) found that organizations that participate in knowledge sharing practices were more likely to predict future market opportunities, improve established processes and create more value for the organization, shareholders and consumers. Numerous authors note that, in order to gain a competitive edge, more companies are including consumer knowledge in the developing of services, products and marketing (Cova & Dalli, 2009; Gibbert et al., 2002; Prahalad & Ramaswamy, 2000). Results further show that the knowledge which organizations extract from consumers often results in more desirable outcomes than if the organization had worked independently (Ramirez, 1999).

However, knowledge sharing is beneficial not only to industry, but also to higher education. As Spohrer and Maglio (2008) note, higher education often adopts a traditional manufacturing mindset, where consumer roles are given less information or power to change the production process. Higher education as a traditional organization, therefore, may be less flexible and slower to change than other kinds of organizations. Yet, as the higher education market grows more competitive, user resource integration may help innovate institutions. von Hippel (2005) has written that consumers who experience specific problems within their daily lives are the same people who are most suitable to solve those problems. For example, the student who struggled within the higher education system is the same student who might offer the most insightful guidance on how to improve upon it. Cook-Sather, Bovill, and Felten (2014) add that while students are not disciplinary experts, they are experts at being students, and therefore have the ability and knowledge frame necessary to contribute meaningfully to advancement of practice. If institutions allow students to share knowledge more freely, this may help them to innovate their service while avoiding future risks (Spohrer & Maglio, 2008).

Equity

Yet while knowledge sharing is key to facilitating co-production, the process cannot take place without the additional function of access, which allows users of the organization to offer knowledge and resources (Gummesson, 2002). However, while research such as that by Fisher and Smith (2011) have found that balanced relationships and equal access were important factors in the co-creation/co-production process, they also found that most relationships between organizations and consumers are unlikely to meet the expected level of mutual reciprocity. The ability for the organization to provide equal access to consumers may be contingent upon two factors. First, Prahalad and Ramaswamy (2004) write that an organization's ability to provide equity to their consumers is often deeply rooted in the organization's narcissism and unwillingness to relinquish control. Secondly, Payne, Storbacka, and Frow (2008) further found that equity is closely linked to the organization's ability to provide a supportive environment for consumers.

However, regardless of the barriers to providing equity, it is key both for co-production and later for the value co-creation process (Navarro-García et al., 2015; Vargo & Lusch, 2008). Value co-creation, like the educational process, needs to allow any consumers or

students to participate, not only selected lead users (Prahalad & Ramaswamy, 2004). Yet to date, many higher education initiatives for including student perspectives and voices are still limited to only a few lead users (Healey et al., 2014). Kuh (2007) writes that student engagement in particular has suffered due to its limited inclusivity when activities often only involved a small group of students.

It has also been noted that without accessibility, value co-creation cannot adhere to other important attributes such as deep and ongoing dialogue and interaction between consumers and the organization or shared responsibility of roles (Prahalad & Ramaswamy, 2004). The process of value co-creation can only be jointly constructed by consumers and organizations if it is transparent (Leavy, 2012; Navarro-García et al., 2015). This builds an atmosphere of trust and helps further establish authenticity (Leavy, 2012).

Value co-creation transparency and access also help the process of allowing for more balanced roles between organizations and consumers (Vargo, Maglio, & Akaka, 2008), as an unbalanced relationship may negatively affect other factors of co-creation such as transparency and dialogue, which both rely on trust gained through the balancing of roles (Ramaswamy & Ozcan, 2014). If the organization is providing resources such as a platform and specific knowledge of previous production, the consumer can also provide resources such as feedback and novel ideas for innovation (Ramaswamy & Ozcan, 2014). In the higher education context, it is also important for student participation to be balanced among student groups, as without balance, individual students or a subset of the student population may take control.

Interaction

Lastly, interactions between consumers and organizations are critical to facilitate other aspects of co-production and co-creation such as knowledge sharing and equity (Grönroos, 2006; Payne et al., 2008). Scholars such as Kale, Dyer, and Singh (2001) and Vargo et al. (2008) have likened the interactions between organizations and their users as a process akin to alliance building. In value co-creation, organization and consumer interactions are also encouraged to rest on continuous dialogue (Prahalad & Ramaswamy, 2004). This amounts to two fundamental shifts in the traditional producer-consumer relationship. Firstly, consumers can give feedback at any point along the value chain, not only at the end (Prahalad & Ramaswamy, 2004). And secondly, consumers can give feedback at any time, on all elements, not only on a single selected project or situation (Prahalad & Ramaswamy, 2004).

However, co-production is just one part of value co-creation. As Vargo and Lusch (2008) note, co-production, while allowing consumers to integrate their resources in the value chain, is still predominantly controlled by the organization. It is the organization that can decide how, when and to what extent consumers can co-produce with them. Therefore, co-production is not sufficient to describe the process of value co-creation. The discussion also needs to include what happens to products and services (i.e. value propositions) once they have been produced, and how they are *used* by the user or, in this case, student.

Value-in-use

If co-production is the process that occurs between users and organizations during the design and delivery of the value proposition, value-in-use can be thought of as what

happens to the value for both users and organizations once they start using the co-produced service and/or product. Value-in-use (ViU) is a term that originates from service-dominant logic (SD-logic). First introduced by Vargo and Lusch (2004), ViU explains that all products and services are value-free until the consumer imposes value upon them. SD-logic, therefore, emphasizes the intangibles and relationships as key factors in the value creation process. It argues that organizations no longer solely provide value (e.g. through an offered product) but rather organizations and consumers both play active roles in their interactions as a joint process to co-create value (Díaz-Méndez & Gummesson, 2012). Therefore, regardless of whether the organization is offering a product or a service (no distinction in fact needs to be made), the true value of the exchange is the application of the resource by the consumer (Vargo et al., 2008). Vargo et al. (2008) use the analogy of a car to demonstrate this. Before the consumer drives the car, it has no value – only when the consumer combines his/her ability to drive with the need to travel to a location does the car incur value.

ViU is related to how consumers consume or use the value propositions provided for them through their time, location, experiences, and stories (Fisher & Smith, 2011; Vargo & Lusch, 2004). Therefore, because ViU relies heavily on the consumers' usage to derive value, it is especially interesting to apply to the higher education context. This is because higher education is a 'credence' good, whereby the students are unable to know the value of their educational experiences or their degree qualifications until they have graduated and left the higher education environment (van Vught & Westerheijden, 2010).

ViU is integral to co-creation but is still only a part of the co-creation process. As highlighted in the Vargo et al. (2008) analogy, the consumer plays an active role in shaping the value of the car to him or herself, he does not play an active role necessarily in shaping the value of the car to all users or future consumers of the same model of car. Therefore, while the consumer may have bought the car and possibly derived value from it, because of the lack of consumer resource integration in the car's design, the car is not customer-centric designed, nor is it co-produced with the consumer.

However, neither is value co-creation just co-production of a value proposition between an organization and a user. It is also how the organization and the consumer *use* what they have co-produced (Lusch & Vargo, 2006; Prahalad & Ramaswamy, 2004). Ranjan and Read (2016) have uncovered three constructs which inform the ViU construct: experience, personalization, and relationship. These indicators, similar to constructs underpinning co-production, are processes rather than outcomes, that continue to develop and shift over time. We have applied these constructs to higher education as: (1) How does value co-creation impact student experiences within higher education? (2) To what extent can students personalize their higher education value propositions? And finally, (3) How does value co-creation impact student relationships to their higher education institution? Unlike co-production, these questions refer not to how the student has co-produced value propositions within higher education, but rather, how the value of such co-production impacts the value they see or use. Ranjan and Read (2016) additionally refer to the ViU construct as the 'lived or joint reality of use and experience' (Table 2).

Below, each of these constructs (experience, personalization, relationship) are further modified and discussed within the higher education context.

Table 2. Co-creation second order construct Part II: value-in-use.

Second-order construct	First-order construct	Modification to higher education
Value-in-use	Experience	How does value co-creation impact student experiences within higher education?
	Personalization	To what extent can students personalize their higher education value propositions?
	Relationship	How does value co-creation impact student relationships to their higher education institution?

Experience

The first aspect of ViU is the consumer's experience and how their experience relates to the consumer's perceived value of it (Lusch & Vargo, 2006). User experiences with their value propositions are important not only to the organization but also to the consumer. If the experience is positive, the consumer is more likely to form a positive relationship with the service and/or product, or even with the organization (Füller, Hutter, & Faullant, 2011; Prahalad & Ramaswamy, 2003). This will later impact consumers' future behavior with the product or service (Payne et al., 2008).

However, consumers' experiences impact not only the likelihood of the consumer returning but also whether they choose to promote the organization to other users and make future contributions to the development of the service or product (von Hippel & Katz, 2002). Research also shows that consumers themselves benefit from positive experiences, as they can further increase the value for the consumer (Mathwick & Rigdon, 2004).

In higher education literature, numerous scholars discuss the student experience (Kelly, Dollinger, & Coates, 2016; McAlpine, Jazvac-Martek, & Hopwood, 2009). Shah and Richardson (2016) define the student experience as students' learning experiences, regardless of the mode of delivery. However, increasingly the term 'the student experience' is focused on the ways students can actively participate in higher education and even take on consumer-type roles (Furedi, 2010). This has spawned a research stream of its own, some in favor of students taking on more consumer-type roles (e.g. Bowden & D'Alessandro, 2011) and others not (e.g. Furedi, 2010). Yet as the higher education market grows more competitive, student experiences may continue to be an important performance mechanism to understand the quality of the service.

Personalization

ViU includes not only user experiences with the value proposition, but also how they can further personalize the value proposition to their own usage (Ranjan & Read, 2016). Unlike co-production, where the personalization or consumer preferences are integrated at the start of the production process (Grönroos & Voima, 2013; Ramaswamy & Ozcan, 2014), personalization in ViU occurs later in the value chain (Vargo & Lusch, 2008). The value proposition has been created (or perhaps co-produced) and now allows for consumers to additionally alter the usage, meaning, or value (Cova, Dalli, & Zwick, 2011).

Users, whether they be traditional consumers in an industry (Fisher & Smith, 2011) or students within higher education (Healey et al., 2014) have increasingly shown interest in being able to personalize their experiences. Thus, more organizations are acknowledging that many consumers want to be able to alter or reimagine their products or services after their purchase (Prahalad & Ramaswamy, 2000; Ramaswamy & Ozcan, 2014). Muniz

and Schau (2005) explain that consumers in some circumstances can actually piece together their own value propositions, given the raw parts, not in a superficial or predetermined fashion, but to their own individual and unique needs.

In higher education, personalization through ViU would also allow for students to piece together value propositions, such as degrees or courses, within their higher education experience to suit their own needs or desires. Research on personalization in higher education so far has been defined as more similar to customization, such as student usage of e-learning platforms (Ozkan & Koseler, 2009). However, the term is actually more similar to how students can apply the experience of higher education to their own needs, such as flexible education trends like mixed mode delivery courses (Kim & Bonk, 2006) or alternate timetabling options (Burton & Nesbit, 2008).

Relationship

The final component of ViU is the relational construct. While interactions and relationships may imply similar terms, they are distinct (Carù & Cova, 2003). The relationship construct complements the two previous components – experience and personalization – by adding a new layer of complexity, as consumers' overall relationship to their organization, separate from individual experiences, can also impact the consumers' overall perceived value (Bonsu & Darmody, 2008). This relates to the value co-creation process in particular, as often consumers' relationship with the organization is based on their collaborative experiences such as those that could be created through co-creation. Once again, this process of joint creation is a shift away from the traditional producer-consumer relationship (Arvidsson, 2005), where users and organizations are engaged in new forms of reciprocity (Chandler & Vargo, 2011).

In higher education, the student's relationship to their institution can similarly impact their feelings towards their institution and their higher education experiences, including learning (Carini, Kuh, & Klein, 2006). Students' positive relationships with their institutions can also create a community (Zhao & Kuh, 2004) that may later predict future collaborative behaviors. As value co-creation builds underlying aspects of relationship formation, such as continuous dialogue and balanced roles (Prahalad & Ramaswamy, 2004), it may have the unique capability to improve students' relationships with their institutions.

Linking value co-creation to student and institutional anticipated benefits

In order for value co-creation to have a meaningful impact on institutions and students, a model of value co-creation also needs to link to the anticipated benefits for both students and institutions. Theories such as diffusion of innovation (see Rogers, 2003) posit that new strategies or tools often need to have wide appeal to be adopted; thus, by linking co-creation to both major populations within higher education, the process is more likely to gain widespread interest. However, not much research to date has investigated the value to both students and institutions through a value co-creation framework (e.g. Elsharnouby, 2015; Navarro-García et al., 2015). Therefore, another contribution of this value co-creation model for higher education will be the prediction of institutional and student benefits.

It is unknown how the underlying constructs of co-production or value-in-use (i.e. experience, personalization, relationship) will impact the benefits for students and institutions. However, certain key student factors that are often noted as important to students'

overall positive higher education experiences, such as quality interactions, satisfaction and graduate capabilities, may be facilitated by working closely with members of the faculty, administration, or staff (Healey et al., 2014). As co-production requires a level of knowledge exchange, equity and interaction often built on a foundation of the relationships between faculty and staff, it is likely that the process of value co-creation will positively impact students' overall experiences. Meanwhile, dimensions of value-in-use (ViU) may more strongly link to institutional factors, such as student loyalty, university image, and student-institutional identification. This is because ViU has been shown to influence consumers' experiences, personalization with services/products and their relationships to the organization (Vargo & Lusch, 2008). These elements align to institutional relationship marketing outcomes that encourage positive images of the institution that help students build trust and loyalty, as well as identify themselves as members of the institutional community (Helgesen, 2008). However, the benefits for both students and institutions are likely to correlate with both constructs.

Anticipated benefits for students

The anticipated benefits for students through the value co-creation process may include quality interactions, satisfaction, and graduate capabilities. This section will therefore, within each of these three anticipated benefits, explain why these factors are important for higher education to maintain or improve, and also why value co-creation might help institutions achieve it.

Quality interactions

While much research within higher education has stressed the student–faculty relationship (Komarraju, Musulkin, & Bhattacharya, 2010; Yang, Alessandri, & Kinsey, 2008) only a few studies have looked at the nature – or quality – of interactions (Cox & Orehovec, 2007; Pascarella & Terenzini, 2005), and fewer still through the lens of value co-creation.

Quality interactions, as determined by the frequency and depth of student–faculty discussion and relationships, are key for impacting students' overall value of their higher education experience (Tierney, Corwin, & Colyar, 2005). Quality interactions can improve students' mutual respect and understanding of their faculty (Schlesinger et al., 2015) and lead to better retention (Hurtado & Carter, 1997). Previous research has further revealed that student–faculty 'functional interactions' or academic-related interactions that occur outside of the classroom are the most impactful type of interaction (Cox & Orehovec, 2007). Bowden (2011) additionally found that while satisfaction and trust were critical indicators of student loyalty, research often overlooks the important role of affective commitment, which may be a stronger predicator than satisfaction, thus stressing the need for institutions to provide students with quality faculty and/or staff interactions throughout their experience.

With all the research showing benefits from student-institution quality interactions, higher education will likely continually explore how long-term and trusting relationship building can improve student outcomes (Bowden, 2011). Scholars have emphasized the importance of evaluating out-of-classroom contexts of social interactions (Terenzini, Pascarella, & Blimling, 1996). By realizing that the nature of interactions between institution and student, both inside and outside the classroom, is increasingly important, it is possible

that value co-creation, a process that encourages teachers and students to interact both inside and outside the classroom, might have a major impact in higher education in the coming years (Pluijm, 2010).

Quality interactions have the potential to be positively impacted through the value co-creation process. Students who are engaged in quality relationships with their teachers and professors report higher levels of learning as well as valuing their education experiences more (Komarraju et al., 2010; Umbach & Wawrzynski, 2005). Numerous scholars have also identified that by engaging students in their higher education experience, a relationship between faculty and student may help increase the value of higher education (Bowden, 2011; Healey et al., 2014).

Satisfaction

Satisfaction as a construct has been defined in various ways, with no clear consensus on a definition. Oliver and Swan (1989) write that satisfaction can be an evaluative, affective or emotional response. In consumer literature, customer satisfaction, however, is more defined by the likelihood that the consumer will return to the service provider (McDougall & Levesque, 2000). Satisfaction measures also differ, as consumers or students can be asked about their satisfaction during or directly after an experience, or after considerable time has passed.

Satisfaction is often used in higher education studies because of its versatility. It has been shown to positively link to several important indicators in higher education including retention (Tinto, 1975), trust (Michell, Reast, & Lynch, 1998) and loyalty (Schertzer & Schertzer, 2004). And while a few studies have indicated that co-production is not positively related to participant satisfaction (due to increased pressure or stress) (Zwick, Bonsu, & Darmody, 2008), others have shown that co-creation participation may increase consumer value and satisfaction (Dong, Evans, & Zou, 2008; Ranjan & Read, 2016). As higher education research has suggested that institutional responsiveness, communication and access may all be positive influencers on student satisfaction (Douglas, McClelland, & Davies, 2008), it is likely that co-creation, which facilitates many of these factors, would also positively impact student satisfaction. Further, research has also shown that when consumers are refused when they would like to participate in co-creation processes, they are less satisfied and are less likely to repurchase than prior levels (Roggeveen, Tsiros, & Grewal, 2012). It is, therefore, important that as students seek more participatory roles within higher education, they are given the opportunity to do so meaningfully.

Graduate capabilities

Another possible benefit for students through value co-creation is through their graduate capabilities. A graduate capability is a broad term that encompasses skills that students can transfer or use in lifelong learning and their professional lives (Sumsion & Goodfellow, 2004). In recent years, higher education has also become more focused on providing students with graduate capabilities (Sin & McGuigan, 2013). These skills can range vastly and can be difficult to assess (Sin & McGuigan, 2013). Examples of graduate capabilities include writing clearly and effectively, thinking clearly and effectively, solving complex problems, and understanding different social contexts.

Value co-creation may be able to improve graduate capabilities. Research has shown that engaging students as partners through co-creation in curriculum design and/or

research improves learning and teaching practices for both students and faculty (Cook-Sather, 2014; Healey et al., 2014). A student with the ability to participate actively in institutional projects has also been shown to have more agency (Healey et al., 2014) and enthusiasm for learning (Soohoo, 1993) that may impact their graduate capabilities. The process of value co-creation, where students are jointly producing value propositions with their institution, and therefore actively participating as well as taking on leadership roles, may impact their graduate capabilities.

Anticipated benefits for institutions

Part of the appeal of value co-creation, as opposed to student engagement, which is primarily concerned with the benefits only for the student population, is that the process is inherently one of mutual value creation (Vargo et al., 2008) whereby both the student and the institution stand to benefit. For example, institutions are increasingly concerned with their relational marketing (Judson & Taylor, 2014) or how the student perceives the institution. These perceptions, based on the students' relationships with the institution, may have important benefits to the brand of higher education (Helgesen, 2008) and future alumni behaviors (Khanna, Jacob, & Yadav, 2014). As value co-creation helps build relationships between students and faculty members, it may also be able to benefit the institution through numerous factors including student loyalty, university image as perceived by students, and student-university identification (Schlesinger et al., 2015). Moreover, as value co-creation is a process that encourages continuous and quality interactions between students and institutions (Prahalad & Ramaswamy, 2004), it is possible that these important institutional benefits may be amplified through the process of value co-creation. This section therefore will describe each institutional anticipated benefit hypothesized by this research and give justification as to why and/or how value co-creation may affect it.

Student loyalty

Student loyalty is of growing interest within the higher education sector. Loyalty or students' preferences over time for one specific product or service (or brand) (Dick & Basu, 1994) has been a key factor in measuring the likelihood of referral or repurchase intentions (Brown & Mazzarol, 2009), and thus, an important benefit to institutions. Scholars such as Hennig-Thurau, Langer, and Hansen (2001) have shown a correlation between students' perceived service quality and their loyalty to their institution. However, scholars have also debated whether student loyalty can be measured as a one-dimensional factor or whether it is, in fact, a multi-dimensional construct, only measurable over time (Rundle-Thiele, 2005). Other scholars such as Carvalho and de Oliveira Mota (2010) note that by studying student loyalty post-graduation, student responses may become contaminated by their employment outcomes. Therefore, this model will focus on student loyalty as determined by current students, hopefully reflecting a level uninfluenced by later graduate outcomes.

Student loyalty is an important benefit for institutions because it provides the basis of their relationship with their students turned graduates and alumni (Hennig-Thurau et al., 2001). It assumes that students who exhibit loyalty are more likely to return to their institutions for graduate degrees, refer future students and contribute financially as alumni or through job offers to new graduates (Helgesen, 2008). Yet scholars have not delved deeply

into how loyalty can be nurtured through higher education (Brown & Mazzarol, 2009) and studies looking at student loyalty as an outcome of a process such as value co-creation that facilitates trust and joint value creation may help direct future research.

Student trust can be built through students' relationships with their higher education institutions' personnel and management (Carvalho & de Oliveira Mota, 2010). Through a trusting relationship, students are likely to have stronger perceptions of value in their institutions, which through time may enhance loyalty. Carvalho and de Oliveira Mota (2010) found that within higher education, students are likely to have stronger loyalty towards institutions they view as competent at problem solving and taking care of student problems. Value co-creation as a process is built on developing relationships between students and institutions, and as such, may be linked to student trust and loyalty. Moreover, as institutions will learn more about student preferences and opinions through value co-creation, they may be able to better align with students' needs, and from this, potentially benefit from stronger student loyalty and future alumni networks (Helgesen, 2008; Schlesinger, Cervera, & Pérez-Cabañero, 2017), as student loyalty, which has also been shown to continue past graduation, can manifest in more engaged alumni behaviors (Schlesinger et al., 2017). And loyal alumni can help universities obtain financial resources as well as helping them connect with industry partners (Frasquet et al., 2012).

University image

Related to student loyalty, and another important anticipated benefit for institutions from value co-creation, is university image as perceived by students (Alves & Raposo, 2010). University image, or the impression left by the product, service or brand in the minds of the consumer (Nguyen & Leblanc, 2001), is linked to value perception as well as student satisfaction (Brown & Mazzarol, 2009). It is increasingly important in the competitive higher education market (Alves & Raposo, 2010). University image, moreover, is important to measure because it also informs how the public and community feel about the institution. Universities are also concerned about their image, as the image and reputation also impact the appeal of the institution to future students (Sung & Yang, 2008).

Similar to loyalty, however, university image is complex and difficult to measure (Schlesinger et al., 2015; Sung & Yang, 2008). Yet the term is often mixed up with related terms such as identity and reputation. Moreover, many studies have looked at various antecedents to loyalty, such as personality or prestige, which may also impact image (Sung & Yang, 2008).

Value co-creation, through empowering institutional-student relationships, can increase the institution's ability to cater towards the student population and improve student retention (Zwass, 2010). This is especially important in a market-orientated and competitive higher education context because student retention rates can affect not only current but also prospective students who, if knowledgeable of an institution's low retention rates, may select another institution (Helgesen, 2008).

Student-university identification

The third benefit that this model includes for institutions is student-university identification (S-U ID). Originally adapted from Schlesinger et al. (2015) as graduate-university identification (G-U ID), the term is conceptualized through business literature as the

consumer-to-company identification. It refers to how the organizational members relate to the organization and its values or mission (Bhattacharya & Sen, 2003). If the identification is strong, students will see the institution's successes and failures as their own (Schlesinger et al., 2015).

Ashforth and Mael (1989) have previously argued that as organizational members identify with their organizations, they are more likely to support the development of the organization. Affective components, such as pride, are key aspects to developing a positive social identity (Sung & Yang, 2008). Yet while related to other constructs such as loyalty and image, student-university identification may not be directly impacted by the foundational aspects of loyalty, such as strong interpersonal ties. Instead, students' likelihood of identifying with their institution may be more closely linked to how they view the institution's defining characteristics and values (Dutton, Dukerich, & Harquail, 1994).

S-U ID is also closely linked to brand identity literature, which is of growing interest to the competitive higher education market (Bennett & Ali-Choudhury, 2009). Value co-creation has the potential to help even large organization serve individual customers (Pluijm, 2010). When investigating university branding, studies have shown that relationships and personal interaction are necessary for creating quality experiences (Pinar, Trapp, Girard, & Boyt, 2011). Pinar et al. (2011) stress that the educational experience can be thought of as the sum of the many, small instances of interactions and experiences, all of which have the potential to affect the students' overall value. Through a platform of value co-creation, institutions can further investigate student feedback that will help them assess current realities, plan for future challenges, and detect ongoing weaknesses (Zwass, 2010). This improves upon traditional capitalistic structures where customers had only a low ability to impact the service or product, yet had deep and complex needs (von Hippel, 2005).

Institutions with input from diverse populations of students can also engage in market segment identification, which has been shown to be key in newer marketing strategies (Bock, Poole, & Joseph, 2014). By breaking down distinct student populations, institutions with market segmentation can identify unique needs of diverse groups and further understand the sector perhaps better than their competitors (Bock et al., 2014). Moreover, by facilitating value co-creation, organizations can better allow individual consumers to form like-minded groups (Füller et al., 2011). These customer-driven groups can help organizations anticipate patterns of customer evolution (Prahalad & Krishnan, 2008). Navarro-García et al. (2015) additionally found that co-creation activities could improve students' teamwork, efficient use of time, and the ability to find new ideas and solutions.

Indicative model

The above literature review and conceptual framing help to form a new model of value co-creation for the higher education sector. As mentioned, this model includes the dual constructs of value co-creation outlined by Ranjan and Read (2016) as co-production and value-in-use. The model further highlights the formative indicators of these constructs that help clarify the terms. Co-production as a composite variable can be indicated through knowledge (e.g. students and staff exchanging experiences and information), equity, and interaction. Value-in-use, a construct that brings together indicators that measure how the user or student uses the value proposition once it has been offered,

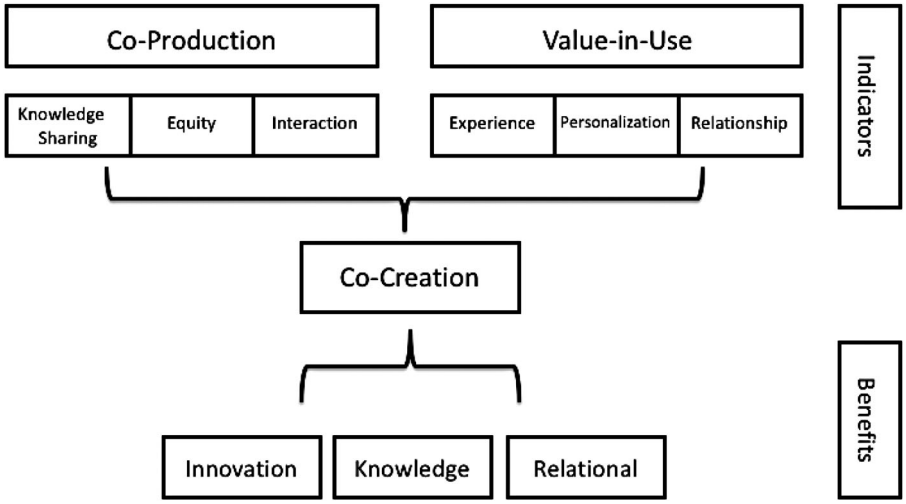

Figure 1. Indicative model of value co-creation in higher education.

can be indicated through experience, personalization and relationship third order constructs. Thus, these six formative indicators, informing the dual constructs of co-production and value-in-use, inform the measure of value co-creation an institution can orientate itself towards (Figure 1).

The model also outlines the anticipated benefits, both for students and institutions, deriving from a value co-creation orientation. Informed by both higher education literature and business literature on value co-creation in other contexts, these anticipated benefits are for students, quality interactions, satisfaction, and graduate capabilities, and for institutions, student loyalty, university image, and student-university identification. The model presented here is indicative and future research should attempt to empirically test this model in authentic situations and environments.

Examples in higher education

Higher education encompasses many activities both in and out of the classroom that may have the potential to be co-created with students. In fact, in some ways, higher education already co-creates with students in small instances, such as eliciting student feedback with questionnaires. Other examples of where student resources are utilized are through student unions or organizations (Varnham, Olliffe, Waite, & Cahill, 2016), designing learning environments (Könings, Bovill, & Woolner, 2017), and students creating university-related content through social media. Emerging research, such as that of Healey et al. (2014) and Bovill, Cook-Sather, Felten, Millard, and Moore-Cherry (2016), highlights how higher education institutions can co-create within the teaching and learning space specifically. However, until research measures these activities through a value co-creation lens (such as the one proposed here), it is not possible to know which activities in higher education provide more co-creation opportunities than others, or even how the presence of co-creation may impact the benefits for both students and institutions.

Discussion

The research presented here not only contributes ongoing conceptualization of value co-creation as a dual construct encompassing both co-production and value-in-use, but it also modifies co-creation to the higher education context. To achieve this, the model both alters the underlying constructs within each dimension of value co-creation to a student-institutional context and predicts factors of value co-creation for both students and institutions.

Now and continually into the future, students will be able to select universities on the basis of which university will provide the value propositions they personally think should be created for them (e.g. employability, research interests, online accessibility) (Prahalad & Ramaswamy, 2004). Education has never been a stand-alone product; it has always been a co-created interaction between the student and the institution, and as such, has a foundation based on relationships. Further, higher education has a history of the inclusion of diverse stakeholders to produce multi-faceted and multi-dimensional outcomes (Kelley, Donnelly, & Skinner, 1990). Higher education, however, is unique from other services in that its value is almost entirely unpredictable until one not only completes the experience but years and even decades later when one uses the experience for knowledge application or career purposes. For this reason, it is important that higher education institutions have a deep understanding of shared responsibility between participants and further emphasize the importance of consumer responsibility to play an active role in shaping the value of their experiences.

However, while research thus far has come a long way in defining value co-creation, applying and identifying co-creation in organizations can be difficult. Value co-creation's principles, such as consumer resource integration, can be abstract in practice and thus difficult to measure (Pluijm, 2010; Prahalad & Ramaswamy, 2004). Yet, while perhaps not practical to measure, value co-creation does have potential to be designed and integrated into higher education systems and processes and, eventually, assessed. Thus, while higher education institutions may not seek to quantify their level of value co-creation, they can, in fact, orientate themselves towards more value co-creation approaches.

Additionally, value co-creation strategy is not an easy undertaking for any organization. Some scholars have even described it as 'service science' where the complex systems must strike a balance to integrate varying resources (Spohrer, Maglio, Bailey, & Gruhl, 2007). Moreover, scholars have posited that value co-creation can only be undertaken in organizations where the firm understands its capabilities and managers are willing to adapt to new principles of co-creation (Pluijm, 2010). Even firms and organizations with explicit value co-creation strategies were found to only superficially understand and exercise co-creation and lacked the motivation to learn from it or develop new insights (Pluijm, 2010). Unsurprisingly, barriers to value co-creation implementation also include a lack of industry leaders acquainted with or knowledgeable on value co-creation. This is troubling, as research shows that leaders are key for institutions hoping to adopt new practices and strategies, including value co-creation strategies (Pucciarelli & Kaplan, 2016). This problem is compounded further in the higher education context as many departmental or sector leaders have backgrounds rooted in academia, not business (Deem & Brehony, 2005). This may lead to unequipped departmental leaders who may be interested in power and influence (Deem & Brehony, 2005), but do not have the background that enables

them to understand core functions of a manager, including executing institutional marketing and management strategies such as value co-creation. Thus, the autonomy that often drives higher education workers into the field can also be the factor that hinders them from adopting more entrepreneurial and managerial attitudes (Pucciarelli & Kaplan, 2016).

Yet the model presented here contributes several benefits to discourse on value co-creation. First, it furthers the work done by scholars such as Ranjan and Read (2016) that argue value co-creation is a nuanced construct, comprising both elements of co-production and value-in-use. It also applies value co-creation to the context of higher education, an increasingly consumer-driven arena where students expect institutions to be capable of change and progress. And finally, the model details specifically how co-creation can add value to higher education, both within the institutions themselves and for the students on whom the institution relies.

Disclosure statement

No potential conflict of interest was reported by the authors.

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References

Alves, H., & Raposo, M. (2010). The influence of university image on student behaviour. *International Journal of Educational Management*, *24*(1), 73–85.

Arvidsson, A. (2005). Brands: A critical perspective. *Journal of Consumer Culture*, *5*(2), 235–258.

Ashforth, B. E., & Mael, F. (1989). Social identity theory and the organization. *Academy of Management Review*, *14*(1), 20–39.

Auh, S., Bell, S. J., McLeod, C. S., & Shih, E. (2007). Co-production and customer loyalty in financial services. *Journal of Retailing*, *83*(3), 359–370.

Bennett, R., & Ali-Choudhury, R. (2009). Prospective students' perceptions of university brands: An empirical study. *Journal of Marketing for Higher Education*, *19*(1), 85–107.

Bhattacharya, C. B., & Sen, S. (2003). Consumer-company identification: A framework for understanding consumers' relationships with companies. *Journal of Marketing*, *67*(2), 76–88.

Bock, D. E., Poole, S. M., & Joseph, M. (2014). Does branding impact student recruitment: A critical evaluation. *Journal of Marketing for Higher Education*, *24*(1), 11–21.

Bonsu, S. K., & Darmody, A. (2008). Co-creating second life: Market—consumer cooperation in contemporary economy. *Journal of Macromarketing*, *28*(4), 355–368.

Bovaird, T., & Loeffler, E. (2013). We're all in this together: Harnessing user and community co-production of public outcomes. *Birmingham: Institute of Local Government Studies: University of Birmingham*, *1*, 15. Retrieved from https://www.birmingham.ac.uk/Documents/college-social-sciences/government-society/inlogov/publications/2013/chapter-4-bovaird-loeffler.pdf

Bovill, C., Cook-Sather, A., Felten, P., Millard, L., & Moore-Cherry, N. (2016). Addressing potential challenges in co-creating learning and teaching: Overcoming resistance, navigating institutional norms and ensuring inclusivity in student–staff partnerships. *Higher Education*, *71*(2), 195–208.

Bovill, C., & Felten, P. (2016). Cultivating student–staff partnerships through research and practice. *International Journal for Academic Development*, *21*(1), 1–3.

Bowden, J. L. H. (2011). Engaging the student as a customer: A relationship marketing approach. *Marketing Education Review*, *21*(3), 211–228.

Bowden, J. L. H., & D'Alessandro, S. (2011). Co-creating value in higher education: The role of interactive classroom response technologies. *Asian Social Science*, *7*(11), 35. Retrieved from http://www.ccsenet.org/journal/index.php/ass/article/view/12839

Brown, R. M., & Mazzarol, T. W. (2009). The importance of institutional image to student satisfaction and loyalty within higher education. *Higher Education*, *58*(1), 81–95.

Burton, S., & Nesbit, P. L. (2008). Block or traditional? An analysis of student choice of teaching format. *Journal of Management & Organization*, *14*(1), 4–19.

Carini, R. M., Kuh, G. D., & Klein, S. P. (2006). Student engagement and student learning: Testing the linkages. *Research in Higher Education*, *47*(1), 1–32.

Carù, A., & Cova, B. (2003). Revisiting consumption experience: A more humble but complete view of the concept. *Marketing Theory*, *3*(2), 267–286.

Carvalho, S. W., & de Oliveira Mota, M. (2010). The role of trust in creating value and student loyalty in relational exchanges between higher education institutions and their students. *Journal of Marketing for Higher Education*, *20*(1), 145–165.

Cefis, E., & Marsili, O. (2005). A matter of life and death: Innovation and firm survival. *Industrial and Corporate Change*, *14*(6), 1167–1192.

Chandler, J. D., & Vargo, S. L. (2011). Contextualization and value-in-context: How context frames exchange. *Marketing Theory*, *11*(1), 35–49.

Chickering, A. W., & Gamson, Z. F. (1999). Development and adaptations of the seven principles for good practice in undergraduate education. *New Directions for Teaching and Learning*, *1999*(80), 75–81.

Cook-Sather, A. (2014). Multiplying perspectives and improving practice: What can happen when undergraduate students collaborate with college faculty to explore teaching and learning. *Instructional Science*, *42*(1), 31–46.

Cook-Sather, A., Bovill, C., & Felten, P. (2014). *Engaging students as partners in learning and teaching: A guide for faculty*. John Wiley & Sons.

Cova, B., & Dalli, D. (2009). Working consumers: The next step in marketing theory? *Marketing Theory*, *9*(3), 315–339.

Cova, B., Dalli, D., & Zwick, D. (2011). Critical perspectives on consumers' role as 'producers': Broadening the debate on value co-creation in marketing processes. *Marketing Theory*, *11*(3), 231–241.

Cox, B. E., & Orehovec, E. (2007). Faculty-student interaction outside the classroom: A typology from a residential college. *The Review of Higher Education*, *30*(4), 343–362.

Czarnitzki, D., & Kraft, K. (2004). Firm leadership and innovative performance: Evidence from seven EU countries. *Small Business Economics*, *22*(5), 325–332.

Deem, R., & Brehony, K. J. (2005). Management as ideology: The case of 'new managerialism' in higher education. *Oxford Review of Education*, *31*(2), 217–235.

Díaz-Méndez, M., & Gummesson, E. (2012). Value co-creation and university teaching quality: Consequences for the european higher education area (EHEA). *Journal of Service Management*, *23*(4), 571–592.

Dick, A. S., & Basu, K. (1994). Customer loyalty: Toward an integrated conceptual framework. *Journal of the Academy of Marketing Science*, *22*(2), 99–113.

Dong, B., Evans, K. R., & Zou, S. (2008). The effects of customer participation in co-created service recovery. *Journal of the Academy of Marketing Science*, *36*(1), 123–137.

Douglas, J., McClelland, R., & Davies, J. (2008). The development of a conceptual model of student satisfaction with their experience in higher education. *Quality Assurance in Education*, *16*(1), 19–35.

Droge, C., Stanko, M. A., & Pollitte, W. A. (2010). Lead users and early adopters on the web: The role of new technology product blogs. *Journal of Product Innovation Management*, *27*(1), 66–82.

Dutton, J. E., Dukerich, J. M., & Harquail, C. V. (1994). Organizational images and member identification. *Administrative Science Quarterly*, *39*, 239–263.

Elsharnouby, T. H. (2015). Student co-creation behavior in higher education: The role of satisfaction with the university experience. *Journal of Marketing for Higher Education*, *25*(2), 238–262.

Fisher, D., & Smith, S. (2011). Cocreation is chaotic: What it means for marketing when no one has control. *Marketing Theory*, *11*(3), 325–350.

Frasquet, M., Calderón, H., & Cervera, A. (2012). University–industry collaboration from a relationship marketing perspective: An empirical analysis in a Spanish University. *Higher Education*, *64*(1), 85–98.

Frow, P., Nenonen, S., Payne, A., & Storbacka, K. (2015). Managing co-creation design: A strategic approach to innovation. *British Journal of Management*, *26*(3), 463–483.

Füller, J., Hutter, K., & Faullant, R. (2011). Why co-creation experience matters? Creative experience and its impact on the quantity and quality of creative contributions. *R&D Management*, *41*(3), 259–273.

Furedi, F. (2010). Introduction to the marketisation of higher education and the student as consumer. *The Marketisation of Higher Education and the Student as Consumer*, 1–8. Retrieved from https://network23.org/freeunisheff/files/2015/07/Mike-Molesworth-Richard-Scullion-Elizabeth-Nixon-The-Marketisation-of-Higher-Education-and-the-Student-as-Consumer-book.pdf

Gibbert, M., Leibold, M., & Probst, G. (2002). Five styles of customer knowledge management, and how smart companies use them to create value. *European Management Journal*, *20*(5), 459–469.

Grönroos, C. (2006). Adopting a service logic for marketing. *Marketing Theory*, *6*(3), 317–333.

Grönroos, C. (2011). Value co-creation in service logic: A critical analysis. *Marketing Theory*, *11*(3), 279–301.

Grönroos, C., & Voima, P. (2013). Critical service logic: Making sense of value creation and co-creation. *Journal of the Academy of Marketing Science*, *41*(2), 133–150.

Gummesson, E. (2002). Relationship marketing in the new economy. *Journal of Relationship Marketing*, *1*(1), 37–57.

Gummesson, E. (2007). Exit services marketing-enter service marketing. *Journal of Customer Behaviour*, *6*(2), 113–141.

Healey, M., Flint, A., & Harrington, K. (2014). *Engagement through partnership: Students as partners in learning and teaching in higher education*. York: HEA.

Helgesen, Ø. (2008). Marketing for higher education: A relationship marketing approach. *Journal of Marketing for Higher Education*, *18*(1), 50–78.

Hennig-Thurau, T., Langer, M. F., & Hansen, U. (2001). Modeling and managing student loyalty: An approach based on the concept of relationship quality. *Journal of Service Research*, *3*(4), 331–344.

Humphreys, A., & Grayson, K. (2008). The intersecting roles of consumer and producer: A critical perspective on co-production, co-creation and prosumption. *Sociology Compass*, *2*(3), 963–980.

Hurtado, S., & Carter, D. F. (1997). Effects of college transition and perceptions of the campus racial climate on Latino college students' sense of belonging. *Sociology of Education*, *70*(4), 324–345.

Judson, K. M., & Taylor, S. A. (2014). Moving from marketization to marketing of higher education: The co-creation of value in higher education. *Higher Education Studies*, *4*(1), 51. Retrieved from http://www.ccsenet.org/journal/index.php/hes/article/view/30222

Kale, P., Dyer, J., & Singh, H. (2001). Value creation and success in strategic alliances: Alliancing skills and the role of alliance structure and systems. *European Management Journal*, *19*(5), 463–471.

Kelley, S. W., Donnelly, J. H., & Skinner, S. J. (1990). Customer participation in service production and delivery. *Journal of Retailing*, *66*(3), 315. Retrieved from https://search.proquest.com/openview/b84b76640bfd52a166073d4c3fdae92f/1?pq-origsite=gscholar&cbl=41988

Kelly P., Dollinger M., & Coates H. (2016). New directions for quality assurance: Transparent outcomes for industry collaboration, research training and student success. *Higher Education Evaluation and Development*, *10*(1), 31–51.

Khanna, M., Jacob, I., & Yadav, N. (2014). Identifying and analyzing touchpoints for building a higher education brand. *Journal of Marketing for Higher Education*, *24*(1), 122–143.

Kim, K. J., & Bonk, C. J. (2006). The future of online teaching and learning in higher education. *Educause Quarterly*, *29*(4), 22–30.

Komarraju, M., Musulkin, S., & Bhattacharya, G. (2010). Role of student–faculty interactions in developing college students' academic self-concept, motivation, and achievement. *Journal of College Student Development*, *51*(3), 332–342.

Könings, K. D., Bovill, C., & Woolner, P. (2017). Towards an interdisciplinary model of practice for participatory building design in education. *European Journal of Education*, *52*(3), 306–317.

Kuh, G. D. (2003). What we're learning about student engagement from NSSE: Benchmarks for effective educational practices. *Change: The Magazine of Higher Learning*, *35*(2), 24–32.
Kuh, G. D. (2007). How to help students achieve. *Chronicle of Higher Education*, *53*(41), B12–B13.
Leavy, B. (2012). Collaborative innovation as the new imperative–design thinking, value co-creation and the power of "pull". *Strategy & Leadership*, *40*(2), 25–34.
Lengnick-Hall, C. A. (1996). Customer contributions to quality: A different view of the customer-oriented firm. *Academy of Management Review*, *21*(3), 791–824.
Lusch, R. F., & Vargo, S. L. (2006). Service-dominant logic: Reactions, reflections and refinements. *Marketing Theory*, *6*(3), 281–288.
Mark, E. (2013). Student satisfaction and the customer focus in higher education. *Journal of Higher Education Policy and Management*, *35*(1), 2–10.
Mathwick, C., & Rigdon, E. (2004). Play, flow, and the online search experience. *Journal of Consumer Research*, *31*(2), 324–332.
McAlpine, L., Jazvac-Martek, M., & Hopwood, N. (2009). Doctoral student experience in education: Activities and difficulties influencing identity development. *International Journal for Researcher Development*, *1*(1), 97–109.
McDougall, G. H., & Levesque, T. (2000). Customer satisfaction with services: Putting perceived value into the equation. *Journal of Services Marketing*, *14*(5), 392–410.
Michell, P., Reast, J., & Lynch, J. (1998). Exploring the foundations of trust. *Journal of Marketing Management*, *14*(1–3), 159–172.
Muniz, A. M., & Schau, H. J. (2005). Religiosity in the abandoned Apple Newton brand community. *Journal of Consumer Research*, *31*(4), 737–747.
Navarro-García, A., Peris-Ortiz, M., & Rueda-Armengot, C. (2015). Value co-creation, collaborative learning and competences in higher education. In M. Peris-Oritiz & J. M. Merigó Lindahl (Eds.), *Sustainable learning in higher education, innovation, technology, and knowledge management* (pp. 37–45). Springer International.
Nguyen, N., & Leblanc, G. (2001). Corporate image and corporate reputation in customers' retention decisions in services. *Journal of Retailing and Consumer Services*, *8*(4), 227–236.
Oliver, R. L., & Swan, J. E. (1989). Equity and disconfirmation perceptions as influences on merchant and product satisfaction. *Journal of Consumer Research*, *16*(3), 372–383.
Ozkan, S., & Koseler, R. (2009). Multi-dimensional students' evaluation of e-learning systems in the higher education context: An empirical investigation. *Computers & Education*, *53*(4), 1285–1296.
Pascarella, E. T., & Terenzini, P. T. (2005). *How college affects students: A third decade of research* (Vol. 2). San Francisco: Jossey-Bass.
Payne, A. F., Storbacka, K., & Frow, P. (2008). Managing the co-creation of value. *Journal of the Academy of Marketing Science*, *36*(1), 83–96.
Perks, H., Gruber, T., & Edvardsson, B. (2012). Co-creation in radical service innovation: A systematic analysis of microlevel processes. *Journal of Product Innovation Management*, *29*(6), 935–951.
Pinar, M., Trapp, P., Girard, T., & Boyt, T. E. (2011). Utilizing the brand ecosystem framework in designing branding strategies for higher education. *International Journal of Educational Management*, *25*(7), 724–739.
Pluijm, L. (2010). *Realizing co-creation* (*Master's thesis*). Tilburg University.
Prahalad, C. K., & Krishnan, M. S. (2008). *The new age of innovation*. McGraw-Hill Professional.
Prahalad, C. K., & Ramaswamy, V. (2000). Co-opting customer competence. *Harvard Business Review*, *78*(1), 79–90.
Prahalad, C. K., & Ramaswamy, V. (2003). The new frontier of experience innovation. *MIT Sloan Management Review*, *44*(4), 12–18.
Prahalad, C. K., & Ramaswamy, V. (2004). Co-creation experiences: The next practice in value creation. *Journal of Interactive Marketing*, *18*(3), 5–14.
Pucciarelli, F., & Kaplan, A. (2016). Competition and strategy in higher education: Managing complexity and uncertainty. *Business Horizons*, *59*(3), 311–320.
Ramaswamy, V., & Ozcan, K. (2014). *The co-creation paradigm*. Stanford University Press.
Ramirez, R. (1999). Value co-production: Intellectual origins and implications for practice and research. *Strategic Management Journal*, *20*, 49–65.

Ranjan, K. R., & Read, S. (2016). Value co-creation: Concept and measurement. *Journal of the Academy of Marketing Science*, *44*(3), 290–315.
Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). New York: Free Press.
Roggeveen, A. L., Tsiros, M., & Grewal, D. (2012). Understanding the co-creation effect: When does collaborating with customers provide a lift to service recovery? *Journal of the Academy of Marketing Science*, *40*(6), 771–790.
Rundle-Thiele, S. (2005). Exploring loyal qualities: Assessing survey-based loyalty measures. *Journal of Services Marketing*, *19*(7), 492–500.
Schertzer, C. B., & Schertzer, S. M. (2004). Student satisfaction and retention: A conceptual model. *Journal of Marketing for Higher Education*, *14*(1), 79–91.
Schlesinger, W., Cervera, A., & Iniesta, MÁ. (2015). Key elements in building relationships in the higher education services context. *Journal of Promotion Management*, *21*(4), 475–491.
Schlesinger, W., Cervera, A., & Pérez-Cabañero, C. (2017). Sticking with your university: The importance of satisfaction, trust, image, and shared values. *Studies in Higher Education*, *42*(12), 2178–2194.
Shah, M., & Richardson, J. T. (2016). Is the enhancement of student experience a strategic priority in Australian universities? *Higher Education Research & Development*, *35*(2), 352–364.
Sin, S., & McGuigan, N. (2013). Fit for purpose: A framework for developing and assessing complex graduate attributes in a changing higher education environment. *Accounting Education*, *22*(6), 522–543.
Soohoo, S. (1993). Students as partners in research and restructuring schools. In *The educational forum* (Vol. 57, No. 4, pp. 386–393). Taylor & Francis Group.
Spohrer, J., & Maglio, P. P. (2008). The emergence of service science: Toward systematic service innovations to accelerate co-creation of value. *Production and Operations Management*, *17*(3), 238–246.
Spohrer, J., Maglio, P. P., Bailey, J., & Gruhl, D. (2007). Steps toward a science of service systems. *Computer*, *40*(1), 71–77.
Sumsion, J., & Goodfellow, J. (2004). Identifying generic skills through curriculum mapping: A critical evaluation. *Higher Education Research & Development*, *23*(3), 329–346.
Sung, M., & Yang, S. U. (2008). Toward the model of university image: The influence of brand personality, external prestige, and reputation. *Journal of Public Relations Research*, *20*(4), 357–376.
Terenzini, P. T., Pascarella, E. T., & Blimling, G. S. (1996). Students' out-of-class experiences and their influence on learning and cognitive development: A literature review. *Journal of College Student Development*, *37*, 149–162.
Tierney, W. G., Corwin, Z. B., & Colyar, J. E. (2005). *Preparing for college: Nine elements of effective outreach*. SUNY Press.
Tinto, V. (1975). Dropout from higher education: A theoretical synthesis of recent research. *Review of Educational Research*, *45*(1), 89–125.
Tomlinson, M. (2008). 'The degree is not enough': Students' perceptions of the role of higher education credentials for graduate work and employability. *British Journal of Sociology of Education*, *29*(1), 49–61.
Tsigkas, A. C. (2013). The lean enterprise. In *The lean enterprise* (pp. 1–10). Berlin: Springer.
Umbach, P. D., & Wawrzynski, M. R. (2005). Faculty do matter: The role of college faculty in student learning and engagement. *Research in Higher Education*, *46*(2), 153–184.
van Vught, F., & Westerheijden, D. F. (2010). Multidimensional ranking. *Higher Education Management and Policy*, *22*(3), 1–26.
Vargo, S. L., & Lusch, R. F. (2004). Evolving to a new dominant logic for marketing. *Journal of Marketing*, *68*(1), 1–17.
Vargo, S. L., & Lusch, R. F. (2008). Service-dominant logic: Continuing the evolution. *Journal of the Academy of Marketing Science*, *36*(1), 1–10.
Vargo, S. L., & Lusch, R. F. (2011). Stepping aside and moving on: A rejoinder to a rejoinder. *European Journal of Marketing*, *45*(7/8), 1319–1321.
Vargo, S. L., Maglio, P. P., & Akaka, M. A. (2008). On value and value co-creation: A service systems and service logic perspective. *European Management Journal*, *26*(3), 145–152.

Varnham, S., Olliffe, B., Waite, K., & Cahill, A. (2016). *Student engagement in university decision-making and governance: Towards a more systemically inclusive student voice*.

Vickers, P., & Bekhradnia, B. (2007). *The economic costs and benefits of international students*. Oxford: Higher Education Policy Institute.

von Hippel, E. (2005). Democratizing innovation: The evolving phenomenon of user innovation. *Journal für Betriebswirtschaft*, *55*(1), 63–78.

von Hippel, E. (2009). Democratizing innovation: The evolving phenomenon of user innovation. *International Journal of Innovation Science*, *1*(1), 29–40.

von Hippel, E., & Katz, R. (2002). Shifting innovation to users via toolkits. *Management Science*, *48*(7), 821–833.

Yang, S. U., Alessandri, S. W., & Kinsey, D. F. (2008). An integrative analysis of reputation and relational quality: A study of university-student relationships. *Journal of Marketing for Higher Education*, *18*(2), 145–170.

Zhao, C. M., & Kuh, G. D. (2004). Adding value: Learning communities and student engagement. *Research in Higher Education*, *45*(2), 115–138.

Zwass, V. (2010). Co-creation: Toward a taxonomy and an integrated research perspective. *International Journal of Electronic Commerce*, *15*(1), 11–48.

Zwick, D., Bonsu, S. K., & Darmody, A. (2008). Putting consumers to work: Co-creation and new marketing govern-mentality. *Journal of Consumer Culture*, *8*(2), 163–196.